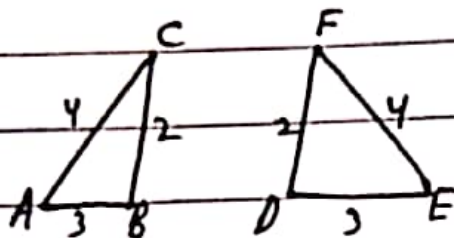
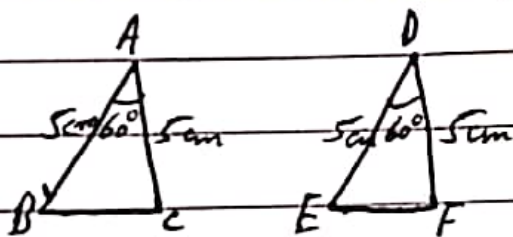


Circles.→ Rules of Congruence:→ Rule 1: SSS - (Side Side Side).

$$AB = DE = 3\text{cm}$$

$$AC = EF = 4\text{cm}$$

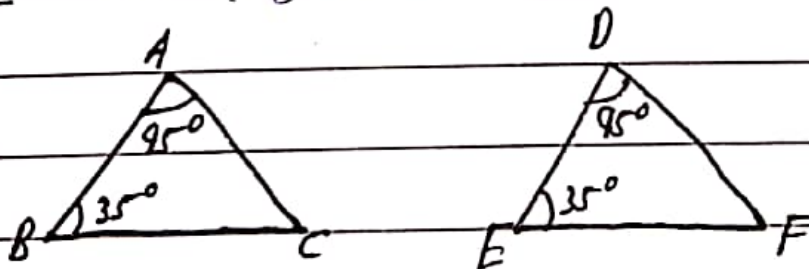
$$BC = DF = 2\text{cm}$$

 $\therefore \triangle ABC$ is congruent to $\triangle DEF$ ($\triangle ABC \cong \triangle DEF$).
→ Rule 2: SAS (Side Angle Side)

$$\angle A = \angle D = 60^\circ$$

$$AB = DE = 5\text{cm}$$

$$AC = DF = 5\text{cm}$$

 $\therefore \triangle ABC \cong \triangle DEF$
→ Rule-3: ASA (Angle side Angle).

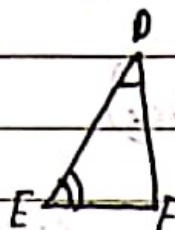
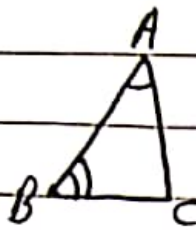
$$\angle A = \angle D = 95^\circ$$

$$\angle B = \angle E = 35^\circ$$

$$AB = DE$$

 $\therefore \triangle ABC \cong \triangle DEF$

→ Rule 4: AAS (Angle Angle Side)



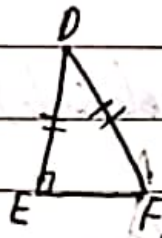
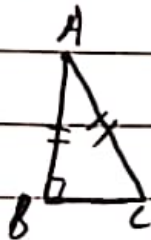
$$\angle A = \angle D$$

$$\angle B = \angle E$$

$$BC = EF$$

$$\triangle ABC \cong \triangle DEF$$

→ RHS Rule: (Right Hypotenuse Side)



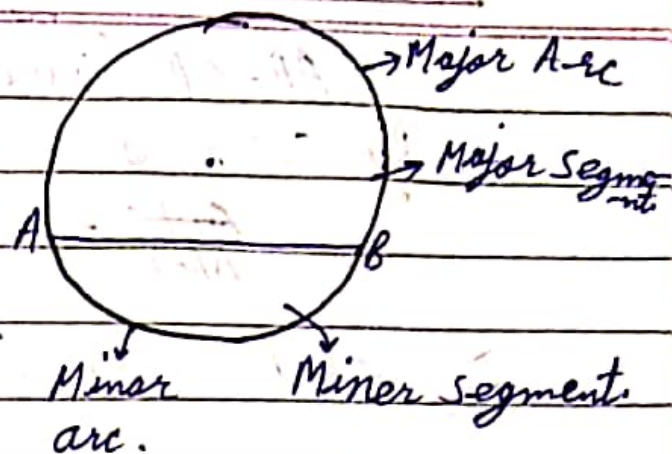
$$AC = DF$$

$$AB = DE$$

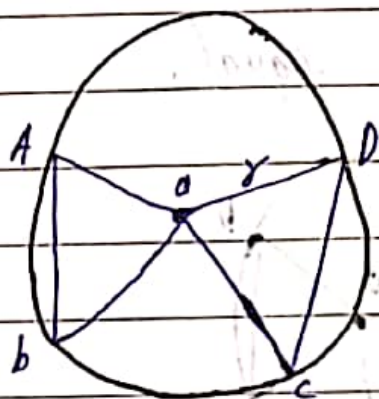
$$\triangle ABC \cong \triangle DEF$$

⇒ Circle - Circle is the infinite set of points equidistant from a fixed point.

→ Chord - A line segment with its end points on the boundary of the circle.



- Theorem 9.1: Equal chords of a circle subtend equal angles at the centre.



- Given: Circle with centre O and radius r , i.e. $C(O, r)$.
Chords AB and CD are such that:
 $AB = CD$.

- To Prove: $\angle AOB = \angle COD$

- Proof: In $\triangle AOB$ and $\triangle COD$

$AO = OD$ (Radii of same circle)

$OB = OC$ (Radii of same circle)

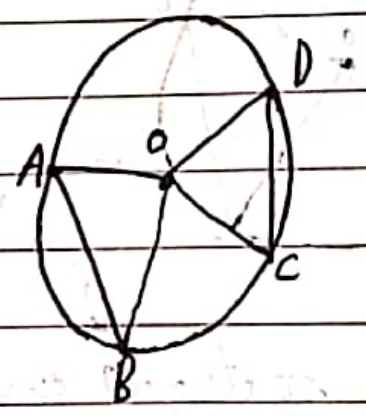
$AB = CD$ (Given)

By SSS congruence

$\triangle AOB \cong \triangle COD$

$\therefore \angle AOB = \angle COD$ (C.P.C.T)

$\Rightarrow$ Theorem 9.2 - If the Angles subtended by the chords of a circle at the centre are equal, then the chords are equal.



$\rightarrow$ Given: A circle with centre O, chords AB and CD, such that,

$\angle AOB = \angle COD$

$\rightarrow$ To Prove - $AB = CD$

$\rightarrow$ Proof - In $\triangle AOB$ and $\triangle COD$.

$AO = OD$ (Radii of the same circle).

$OB = OC$ (Radii of the same circle).

$\angle AOB = \angle COD$ (Given)

$\therefore$ By SAS congruence.

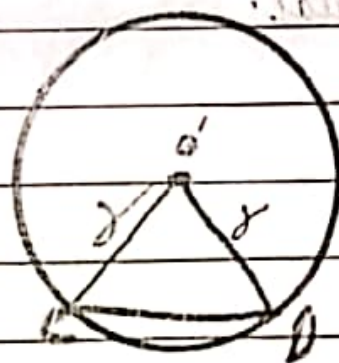
$$\triangle AOB \cong \triangle COD$$

$$AB = CD \text{ (C.P.C.T.)}$$

EXERCISE - 8.1

Q1) Recall that two circles are congruent if they have the same radii.?

Sol.:



Given: Two Circles

$C_1 (O, r)$ and

$C_2 (O', r)$

- Chords AB of C_1 and CD of C_2 are, such that.
 $AB = CD$.

→ To prove - $\angle AOB = \angle CO'D$

- Proof - In $\triangle AOB$ and $\triangle CO'D$
 $AO = CO' = r$ (Circles are congruent)
 $BO = DO = r$ (Circles are congruent)
 $AB = CD$ (Given)

$\therefore$ By SSS Congruence

$$\triangle AOB \cong \triangle CO'D$$

$$\angle AOB = \angle CO'D \text{ (C.P.C.T.)}$$

Q2) Prove that if chords of congruent circles subtend equal angles at their centres, then the chords are equal.

Sol:



- Given - Two circles.

Circle 1 (O', r)

Circle 2 (O, r)

$$\therefore \angle CO'D = \angle AOB$$

- To prove: $CD = AB$

- Proof: In $\triangle AOB$ and $\triangle CO'D$

$$\angle CO'D = \angle AOB \text{ (Given)}$$

$$CO' = AO = r$$

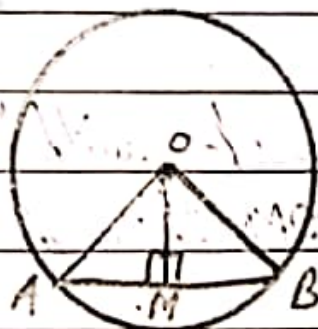
$$BO = DO' = r$$

$\Rightarrow$ By ~~SAS~~ SAS Congruence

$$\triangle AOB \cong \triangle CO'D$$

$$AB = CD \text{ (C.P.C.T)}$$

$\Rightarrow$ Theorem 9.3 - The Perpendicular from the centre of a circle to a chord bisects the chord.

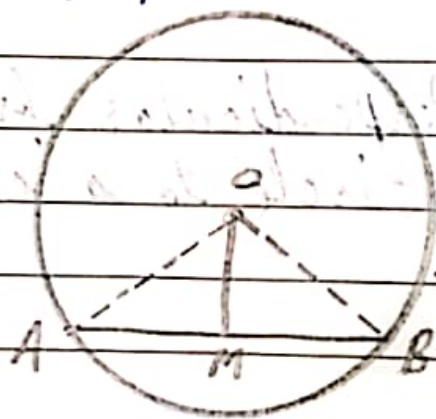


- Given: A circle with centre O, Chord AB and $OM \perp AB$.

- To prove: $AM = MB$

- Const. Join OA and OB
- Proof. In $\triangle AMO$ and $\triangle BMO$
 - $OA = OB$ (Radii of same circle)
 - $OM = OM$ (Common)
 - $\angle AMO = \angle BMO$ (Each 90°)
- $\Rightarrow$ By SSA Congruence
 - $\triangle AMO \cong \triangle BMO$
 - $AM = BM$ (C.P.C.T)

$\Rightarrow$ Theorem (9.4): A line drawn from the centre of the circle to bisect the chord is perpendicular to the chord.



- Given: A circle with centre O , AB is chord and line OM through centre O , such that $AM = MB$
- To prove: $OM \perp AB$
- Const. Join AO and BO .

- Proof- In $\triangle AMO$ and $\triangle BMO$
 $AO = BO$ (Radii of same circle)
 $OM = OM$ (Common)
 $AM = MB$ (Given)

- By SSS congruence

$$\triangle AMO \cong \triangle BMO$$

$$\angle AMO = \angle BMO = x.$$

- Now, $\angle AMO + \angle BMO = 180^\circ$

$$x + x = 180^\circ$$

$$2x = 180^\circ$$

$$x = \frac{180^\circ}{2} = 90^\circ$$

$$\therefore \angle AMO = \angle BMO = x = 90^\circ$$

$$\Rightarrow OM \perp AB.$$

$\Rightarrow$ Theorem (9.5) Equal chords of circle (or congruent circle) are equidistant from the centre.



- Given- A circle with centre 'O'. AB and CD are

two chords of the same circle

$$AB = CD$$

$$OM \perp AB$$

$$ON \perp CD$$

- To prove $OM = ON$

- Constr. Join AO and OB .

- Proof: Since $OM \perp AB \Rightarrow AM = MB$ --- ①

$$ON \perp CD \Rightarrow CN = ND$$
 --- ②

$$AB = CD \text{ (Given)}$$

$$AM + MB = CN + ND$$

$$AM + AM = ND + ND$$

$$2AM = 2ND$$

$$\boxed{AM = ND}$$
 --- ③

- In $\triangle AMO$ and $\triangle DNO$

$$AO = DO \text{ (Equal Radii)}$$

$$AM = ND \text{ (Proved)}$$

$\therefore$ By RHS congruence

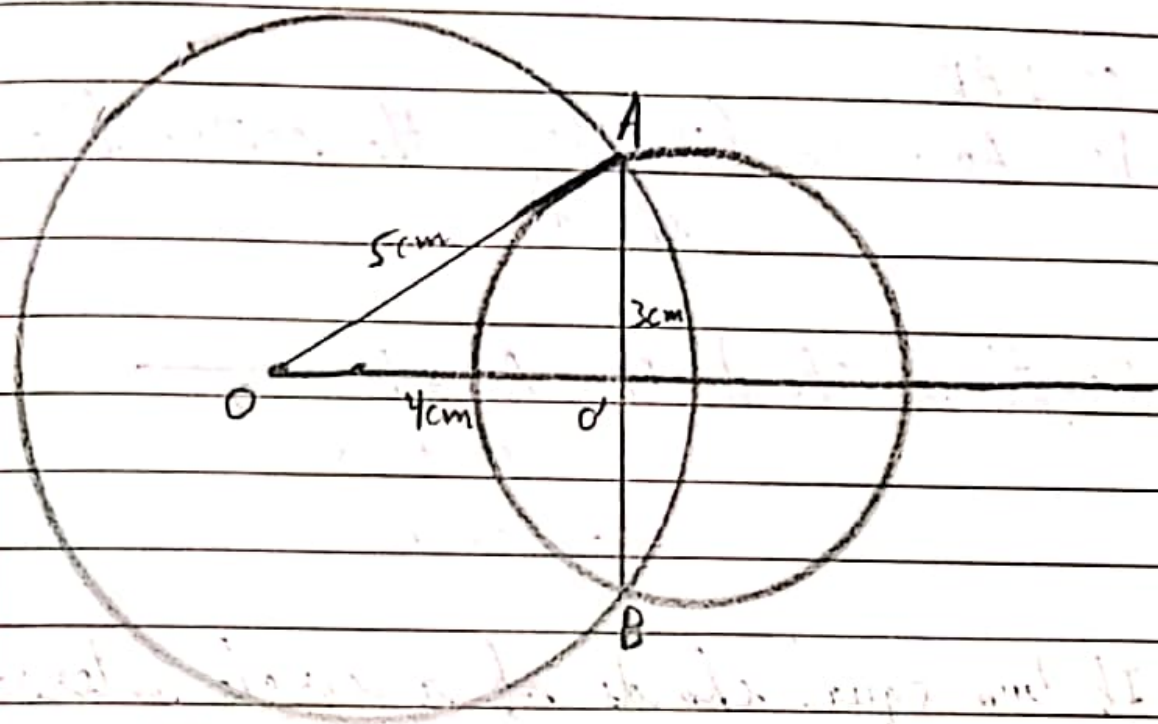
$$\triangle AMO \cong \triangle DNO$$

$$\Rightarrow OM = ON \text{ (C.P.C.T.)}$$

Exercise = 8.2

Q1:- Two circles of radii 5cm and 3cm intersect at two points and the distance between their centres is 4cm. Find the length of common chord.

Soln



- Given: Two circles with centre O and O',
Radii 5cm and 3cm. Such that,
 $OO' = 4\text{cm}$

- Const- Join OA.

- In $\triangle OO'A$

$$(OA)^2 = 5^2 = 25 \text{ --- (1)}$$

$$(OO')^2 = O'A^2 = 4^2 + 3^2 = 16 + 9$$

$$= 25 \text{ --- (2)}$$

- From (1) and (2), we have: $OA^2 = (OO')^2 + O'A^2$

$$\text{i.e., } H^2 = B^2 + P^2$$

$\Rightarrow \Delta OO'A$ is a right angled Δ .

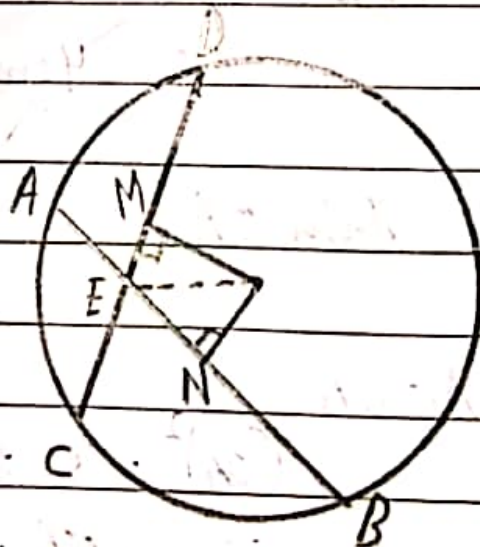
$$\Rightarrow OO' \perp AB$$

$\Rightarrow BO' = AO' = 3\text{cm}$ ($\because$ $\perp$ from the centre bisects the chord).

$$\begin{aligned} \therefore \text{Length of Chord } AB &= AO' + BO' \\ &= 3\text{cm} + 3\text{cm} \\ &= 6\text{cm}. \end{aligned}$$

Q2~ If two equal chords of a circle intersect within the circle, ----?

Sol~



- Given- A circle with centre 'O' chords AB and

CD intersect at point E. and $AB = CD$.

- To Prove- $ED = EB$
 $CE = AE$.

- Const. 1) Draw $ON \perp AB$, $OM \perp CD$
2) Join EO.

- Proof- In $\triangle EMO$ and $\triangle ENO$.
 $\angle M = \angle N$ (Each 90°)
 $OE = OE$ (Common)
 $OM = ON$ (Chords Equidistant)

- By RHS congruence.

$$\triangle EMO \cong \triangle ENO.$$

$$\Rightarrow EN = EM \text{ (C.P.C.T)} \dots \textcircled{1}$$

- Since $AN = \frac{1}{2}AB \dots \textcircled{2}$ ($\because N$ is mid-point of AB)

$$\text{and } CM = \frac{1}{2}CD.$$

$$\Rightarrow AB = 2AN \dots \textcircled{3}$$

$$\Rightarrow CD = 2CM \dots \textcircled{4}$$

- Since $AB = CD$ (Given)

$$2AN = 2CM \text{ (by 3, 4)}$$

$$\Rightarrow AN = CM \Rightarrow \boxed{CM = AN} \dots \textcircled{5}$$

- Subtracting (1) from (5)

$$CM - EN = AN - EN$$

$$CE = AE$$

$$\Rightarrow \boxed{EC = EA} \text{ --- (6)}$$

- Again $CD = AB$ --- (7) (Given)

$\Rightarrow$ Subtracting (6) from (7)

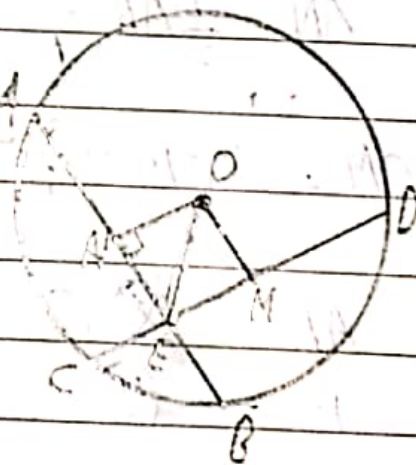
$$CD - EC = AB - EA$$

$$\boxed{ED = EB}$$

Solved
29/08/2026

Q3+ If two equal chords of a circle intersect within the circle,?

Sol)



- Given - Two equal chords of a circle intersect

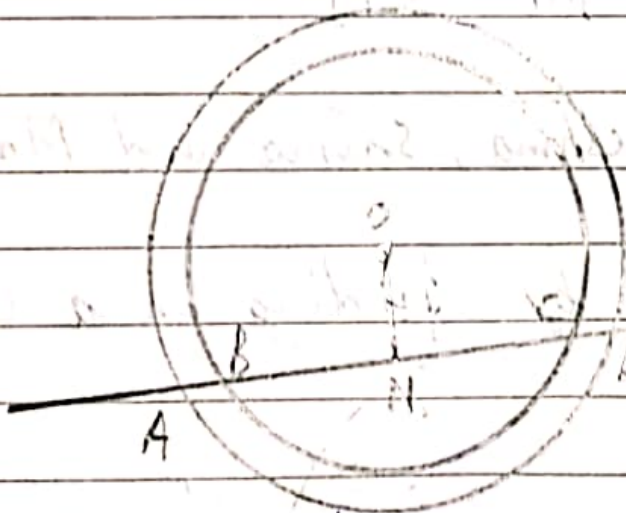
within the circle.

- Proof - In $\triangle OEN$ and $\triangle OEM$
 $OE = OE$ (Common)
 $\angle N = \angle M$ (Each 90°)
 $ON = OM$

- By RHS congruence.
 $\triangle OEN \cong \triangle OEM$
 $\angle 1 = \angle 2$ (C.P.C.T)

Q4) If a line intersects two concentric circles with centre O at A, B, C and D, prove that $AB = CD$.

Sol.)



Two concentric

- Given - Two circles with centre O and point A, B, C and D.

- To Prove - $AB = CD$.

Proof - $ON \perp BC$

$\Rightarrow$ Acc. to Theorem (9.3) i.e; The line drawn from the centre of the circle to the chord bisects the chord.

$$\Rightarrow \cancel{BN} BN = NC \text{ --- (1)}$$

$\Rightarrow$ Now in Bigger circle: $ON \perp AD$

- By Theorem (9.3)

$$\Rightarrow AN = ND \text{ --- (2)}$$

$\Rightarrow$ subtract 1 from 2

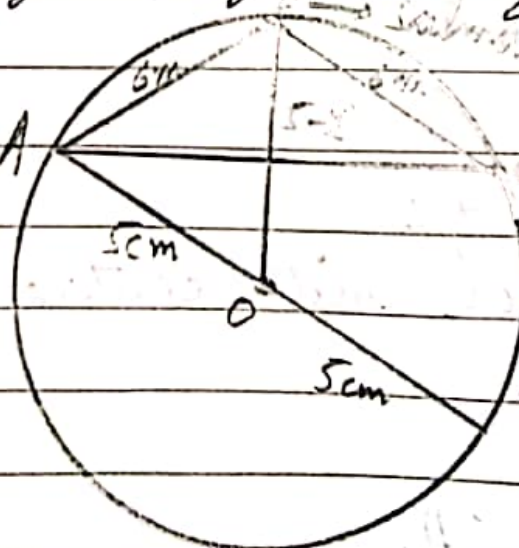
$$\Rightarrow AN - BN = ND - NC$$

$$\boxed{AB = CD}$$

Q5) Three girls Reshma, Salma and Mandip are playing a game by standing on a circle --?

sol)

Reshma - A



Mandip

- Let the position of Reshma, Salma and Mandip be represented by the point A, B and C.

- $AB = BC = 6m$

- Const. Draw $OM \perp BC$
 $OB = 5m$, Let $OM = x$

$\Rightarrow MB = OB - OM$

$MB = 5 - x$

$\Rightarrow$ In right Angled $\triangle AMB$.

- $AB^2 = AM^2 + MB^2$

$6^2 = AM^2 + (5 - x)^2 \quad \dots \textcircled{1}$

- Again, In right Angle $\triangle AOM$

$AO^2 = OM^2 + AM^2$

$5^2 = x^2 + AM^2$

$\Rightarrow AM^2 = 5^2 - x^2 \quad \dots \textcircled{2}$

- From 1 and 2, We have.

$6^2 - (5 - x)^2 = 5^2 - x^2$

$$36 - (5^2 + x^2 - 2(5)(x)) = 5^2 - x^2$$

$$36 - 5^2 - x^2 + 10x$$

$$10x = 5^2 - x^2 - 36 + 5^2 + x$$

$$10x = 2 \times 5^2 - 36$$

$$10x = 2 \times 25 - 36$$

$$10x = 50 - 36$$

$$10x = 14$$

$$x = \frac{14}{10}$$

$$\boxed{x = \frac{7}{5}}$$

- Using Value of x in (2)

$$AM^2 = 5^2 - x^2$$

$$= 5^2 - \left(\frac{7}{5}\right)^2$$

$$= \left(5 + \frac{7}{5}\right) \left(5 - \frac{7}{5}\right)$$

$$AM^2 = \frac{32}{5} \times \frac{18}{5}$$

$$AM^2 = \frac{576}{25}$$

$$AM = \frac{\sqrt{576}}{25}$$

$$AM = \frac{24}{5}$$

$$AM = 4.8 \text{ m.}$$

⇒ Since M is the mid point of AC. ($\because OM \perp AC$)

$$AC = 2AM$$

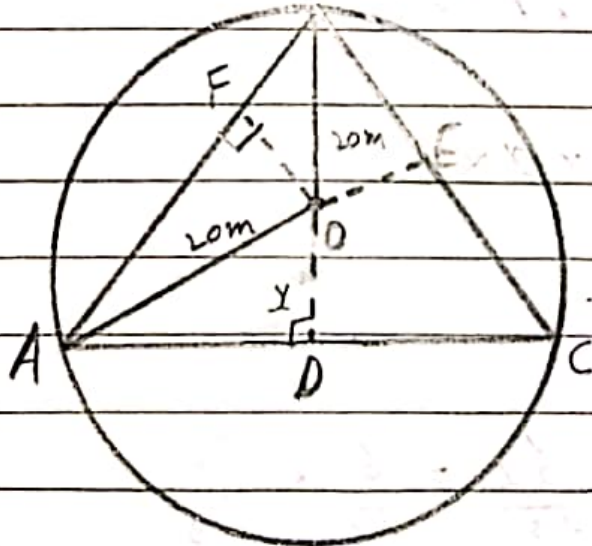
$$AC = 2 \times 4.8$$

$$AC = 9.6 \text{ m}$$

∴ Distance between Salma and Mandip = 9.6 m.

Q6) A Circular park of radius 20m is situated in a colony. The Three boys Ankur, Syed and David are sitting at ^{equal} distance....?

Sol



Let the positions of three boys Ankur, Syed and David be A, B, and C.

- Join OA, OB and OC
- Draw $OD \perp AC$, $OE \perp BC$ and $OF \perp AB$
- Length of $OD = OE = OF = x$
- Since $\triangle ABC$ is an equilateral triangle.
- Let 'a' be the side of the $\triangle ABC$

$$\Rightarrow AB = BC = AC = a$$

$$\text{Area of } \triangle AOC = \text{Area of } \triangle BOC = \text{Area of } \triangle AOB$$

$$\Rightarrow \text{Area of } \triangle ABC = 3 \times \text{Area of } \triangle AOC$$

$$\boxed{\text{Area of equilateral } \triangle = \frac{\sqrt{3}}{4} \cdot \text{side}^2}$$

$$\Rightarrow \frac{\sqrt{3}}{4} a^2 = 3 \times \frac{1}{2} \times AC \times OD$$

$$\frac{\sqrt{3}}{4} a^2 = 3 \times \frac{1}{2} \times a \times x$$

$$\frac{\sqrt{3}}{4} a = \frac{3}{2} \times x$$

$$a = \frac{3x \times \frac{4}{\sqrt{3}}}{\cancel{x} \cdot \sqrt{3}} = \frac{6x}{\sqrt{3}} = \frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} x$$

$$= \frac{6\sqrt{3}x}{3}$$

- Now in right Angle $\triangle ODC$

$$OD^2 + DC^2 = OC^2$$

$$x^2 + \left(\frac{1}{2}AC\right)^2 = 20^2$$